



RAFAEL NAVARRO

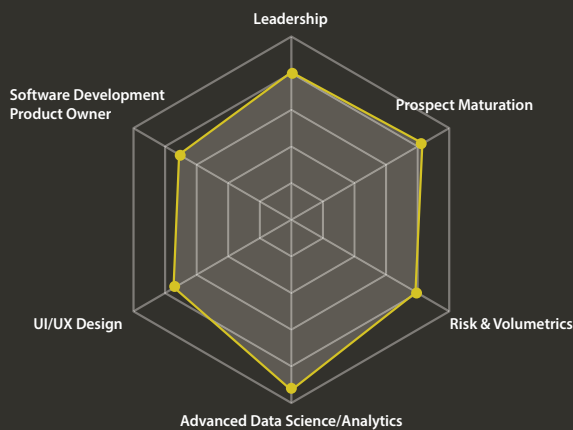
Senior Geoscientist | Subsurface Data Analytics & AI | Product Owner
+15 yrs experience

About Me:

Subsurface domain expert specializing in AI-augmented analytics and product delivery, grounded in seismic-to-well integration, volumetrics/risk, and petrophysical interpretation. I combine interpretation, prospect maturation, and QC rigor with agentic workflows (retrieval + orchestration + automation) to reduce manual effort and increase decision consistency. I deliver maintainable solutions built in VS Code packaged as intuitive advanced visual analytics for broad adoption.

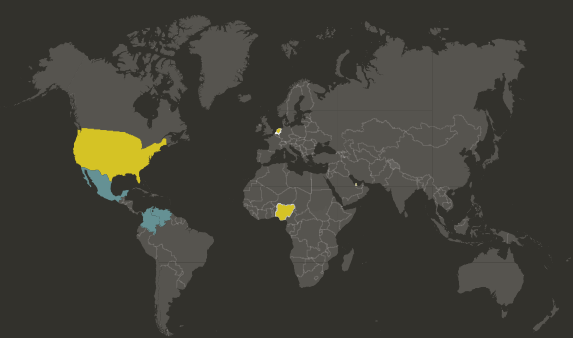
Experienced in supporting WRFM and subsurface opportunity maturation by integrating multi-disciplinary data into structured, decision-ready frameworks that accelerate identification, prioritization, and execution readiness.

Overview



Core Stack: Spotfire (CAP), Databricks (SQL/Python), NLP/LLM workflows, Agentic AI, Figma, Azure DevOps, ArcGIS, VS Code, GitHub.
Domain: Seismic Interpretation, Prospect Maturation, Volumetrics & risk, Petrophysical Interpretation Workflows, WRFM principles.

Certificates: TCA-TIBCO Spotfire®





JUN 2025 - PRESENT

Shell US Hosting Company | Senior Geoscientist Analyst Specialist

Senior Geoscience Analyst focused on scaling subsurface decision-making through advanced analytics, governed workflows, and enterprise-ready data products. I design and deploy interactive dashboards and integrations that connect subsurface users to trusted datasets and accelerate exploration and reservoir evaluation.

- Designed and delivered interactive analytics dashboards enabling faster, more consistent decisions across exploration and subsurface workflows.
- Served as Spotfire Customer Advisor Program (CAP) representative; created new geoscience analytics use cases and drove adoption.
- Built and validated custom data connectors integrating Spotfire with ESRI, Oracle DB, Databricks, and Denodo, including sandbox performance/scalability testing.
- Prototyped AI/NLP (LLM/MCP) concepts for natural-language interaction and automated insight workflows to reduce friction for non-technical users.
- Designed workflows leveraging relational and distributed databases to strengthen data governance and accelerate insight generation across teams.
- Developed solutions including Yet2Find Opportunities (volumetrics), Historical GoA Production (trend analytics), BPA2 QC (volumetrics QC), and PSI petrophysical workflows supporting reservoir evaluation and reservoir-seal pair screening.
- Supported LTRO and WRFM initiatives by enabling development of integrated Spotfire workflows to identify, rank, and prioritize opportunities in waterflood systems, including Gap-to-Potential diagnostics (~\$1B scale).
- Contributed to multidisciplinary subsurface integration through co-visualization of injection, production, and reservoir data, advising on tool capabilities and helping accelerate ISCM diagnostic decision-making.
- Contributing to Venezuela (Monagas) analytics initiatives by harmonizing production and well data, developing decline curve analysis (DCA) workflows (ongoing) to support production diagnostics and opportunity identification in mature onshore assets.



JUN 2020 - FEB 2025

Shell US Hosting Company | Senior X-Digi Geoscientist (Product Owner)

Product Owner bridging geoscience expertise with digital product delivery, leading cross-functional developers and designers to build secure exploration applications. I drove product vision, roadmap priorities, and PoC-led experimentation to validate value early and scale what works.

- Owned product vision, roadmap, and backlog across multiple initiatives; aligned delivery to stakeholder and end-user needs.
- Led cross-functional teams (dev + UX) to deliver secure, impactful exploration applications using rapid PoCs and iterative delivery.
- Integrated technical, commercial, and economic datasets to support opportunity screening and improve decision transparency.
- Leveraged AI and NLP approaches on large subsurface datasets to democratize insights and reduce manual screening effort.
- XStation: Geospatial platform enabling global exploration teams to access integrated datasets and dashboards tailored to user needs.
- ARMPlay: Tools supporting Corporate ARM framework (Attractiveness, Resilience, Materiality) to evaluate and compare opportunities— even with incomplete economics.



OCT 2018 - JUN 2020

Shell International Exploration & Production | **Senior Exploration Geoscientist (Technical Lead)**

Technical lead focused on revitalizing the Nigeria Shallow Water portfolio and scaling decision workflows through digital transformation. I combined play-based exploration leadership with analytics products that improved screening consistency, execution speed, and decision quality.

- Led Nigeria Shallow Water portfolio revitalization in two phases: (1) regional play maps, (2) volume estimates and risk assessments for portfolio refresh.
- Coordinated a multidisciplinary team of ~10 (geology, geophysics, basin modeling, data engineering) to deliver integrated technical outcomes.
- Built the Hydrocarbon Distribution Dashboard enabling penetration charts, fluid distributions, and reservoir properties across 90,000+ reservoirs and 5,000 wells.
- Developed streamlined prospect maturation workflows enabling evaluation of multiple reservoir–seal pairs (RSPs) without compromising QC/Technical quality



JULY 2014 - OCT 2018

Shell Petroleum Development Company of Nigeria (SPDC) | **Exploration Geoscientist**

Exploration geoscientist delivering high-pressure well execution support, subsurface model updates, and exploration-to-development handovers. I focused on operational decision support during drilling and on unlocking value through cost-efficient resource additions.

- Prospect owner for high-pressure wells (e.g., Gbaran 26 & 27), managing complex geology (pressure >0.94) and deep targets (>16,000 ft).
- Provided daily geological updates and operational support; managed VSP acquisition including Look-Ahead VSP techniques for pressure estimation ahead of drilling.
- Updated subsurface model; delivered AARs and results to leadership; performed evaluations (velocity models, volumetrics, QIs, up-dip prospectivity) and executed formal E2D handover.
- Contributed to adding Contingent Resources without additional exploration cost using new technologies/workflows (CTD, AVD, Syn2D Modeling).
- Contributed to 22.8 mmboe handover to Development in 2017; work represented ~15% of Nigeria's CR target.
- Took over HD-JK project to identify drilling opportunities that positively influenced FIDs by adding resources without significant incremental exploration cost.



APRIL 2012 - JULY 2014

Qatar Shell Service Company (QSSC) | **Senior Seismic Interpreter**

Senior seismic interpreter focused on processing integration, well-seismic calibration, and VSP planning/operations to improve subsurface model accuracy and reduce drilling risk. Delivered both technical interpretation outcomes and high-visibility technology trial support.

- Led processing of seismic data collected in Qatar (2012–2013) and coordinated virtually with the Netherlands processing team to improve efficiency and incorporate geological insights into geophysical analysis.
- Responsible for well ties, synthetic generation, and target depth (TD) functions; prepared and managed VSP operations for exploration well QSD-1.
- Provided real-time operational support for VSP operations in collaboration with Qatar Petroleum (North Field appraisal wells).
- Supported novel technology trial: battery-less fiber optic seismic system (~4,000 channels) developed by Shell and PGS; processed data and showcased results to senior stakeholders at a QSRTC event

JUN 2007 - DEC 2012

Exploration Geoscientist | Early Career (Consulting / Mexico / Colombia / Venezuela)

Built a strong technical foundation across seismic interpretation, acquisition QC, velocity modeling, and prospect generation in multiple basins and operating contexts. Delivered integrated geophysical work from field acquisition quality to subsurface mapping and drilling proposals.

- Conducted structural/stratigraphic interpretation of 2D/3D seismic; generated prospects using attributes and detailed mapping across multiple basins.
- Built velocity models and time-depth conversions; tied well and seismic using synthetics and VSP for accurate depth-contoured structural maps.
- Designed and managed a 3D seismic acquisition survey (~60 km²) using Mesa; supported well proposals leading to discoveries and early production uplift.
- Supported seismic QC and re-processing proposals; built velocity models using offset well data for pore pressure prediction and drilling safety support.
- Deepwater support (PEMEX): interpretation/QC workflows and campaign support (e.g., Trion-1 and Supremus-1).
- Seismic acquisition field QC (2D 3C; dynamite sources), field pre-processing, and integrated geophysical methods QC (AMT, refraction, gravity, magnetics, uphole).

Selected Leadership & Hobbies

Led an environmental pollution awareness campaign using real-time public data to increase transparency and community engagement. Within months, the initiative reached thousands of users and gained regional/international media visibility (Undark, IQAir, France24).

- <https://www.iqair.com/blog/success-stories/air-quality-in-africa>
- <https://undark.org/2018/10/22/air-pollution-lagos>
- <https://www.france24.com/en/20180423-nigerians-demand-air-quality-data-over-pollution-fears>

Competitive pickleball (semi-pro track), applied AI and energy innovation, cooking, endurance sports, and family-centered lifestyle



Technology



Energy Optimization



Cooking



Pickleball